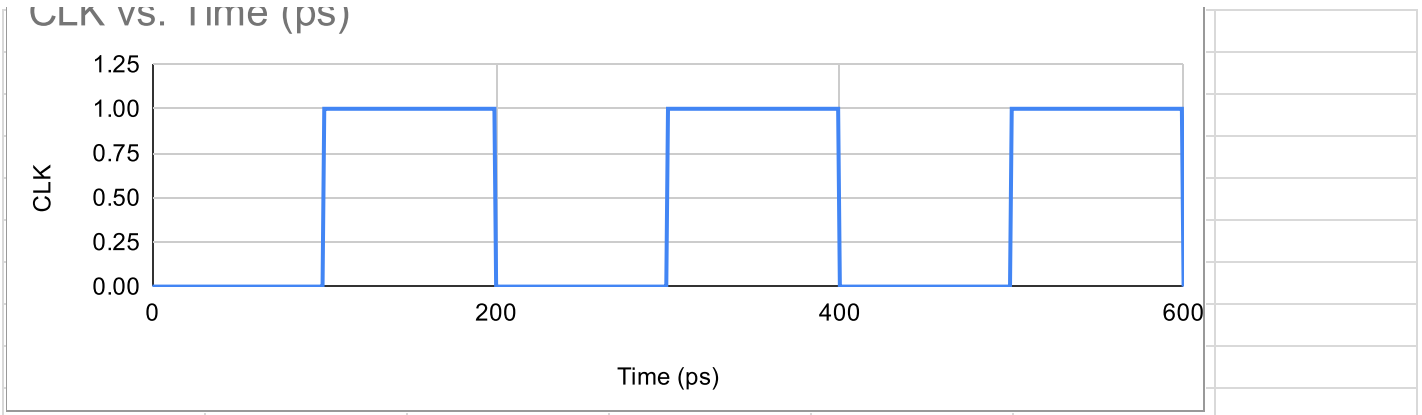
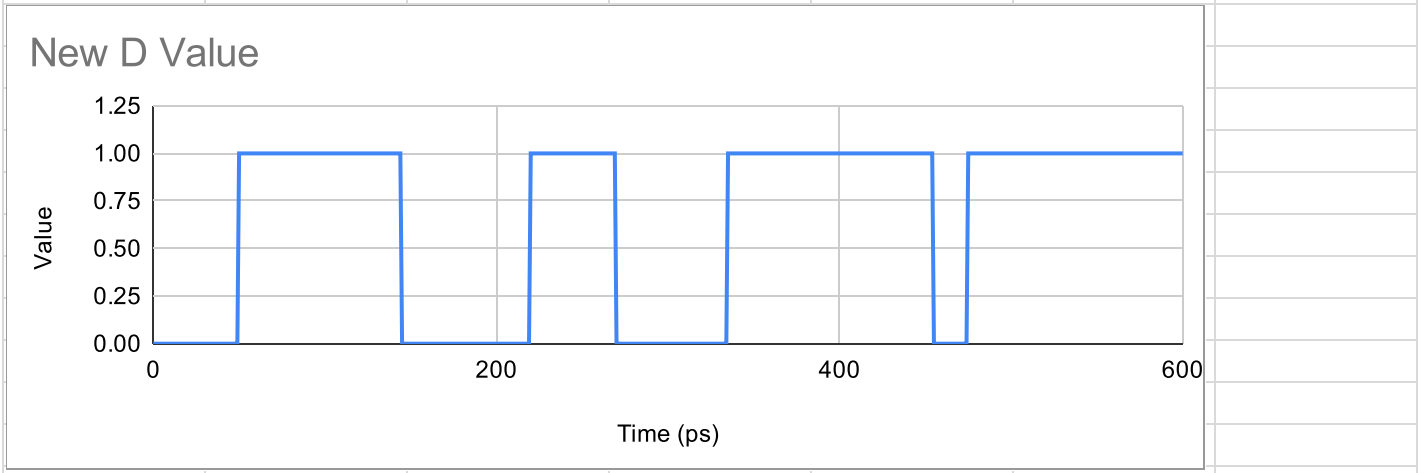
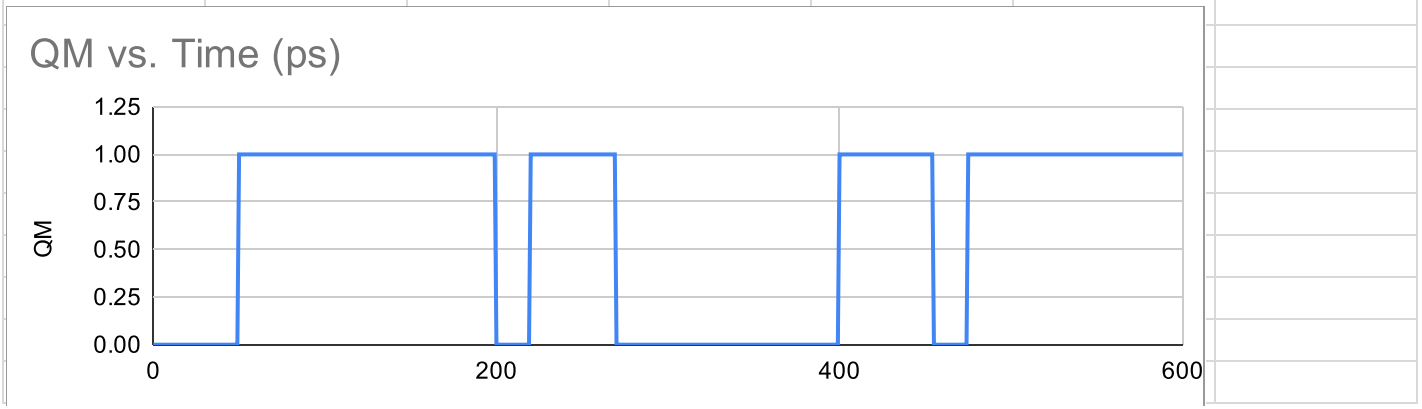
3A

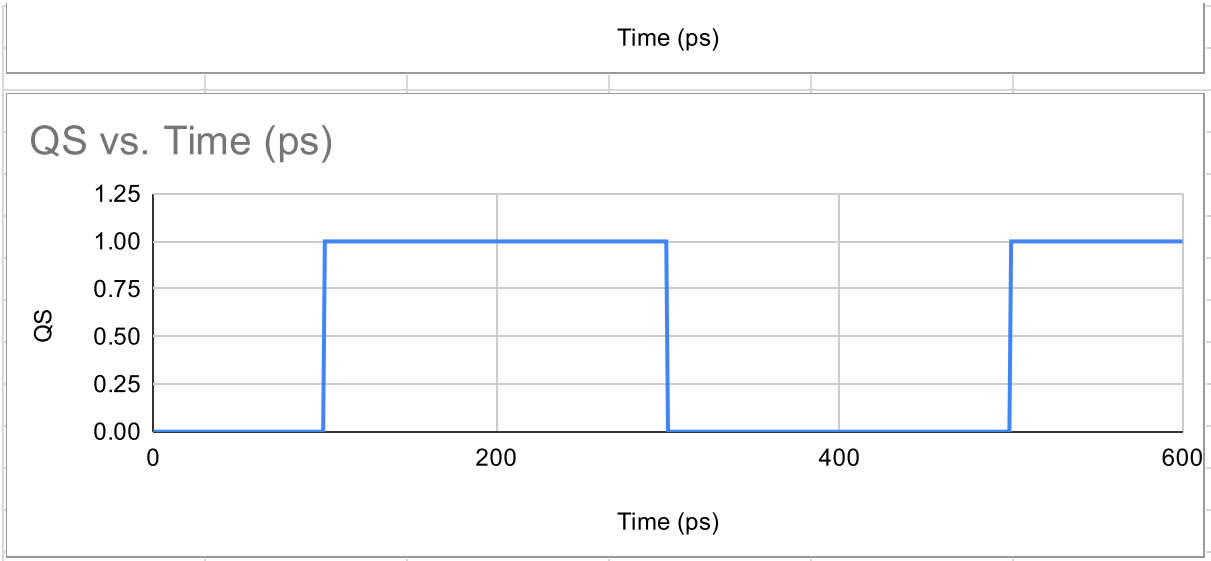
Time (ps)	CLK	New D Value	QM	QS
0	0	0	0	0
49	0	**0**	0	0
50	0	**1**	1	0
99	**0**	1	1	0
100	**1**	1	1	1
144	1	**1**	1	1
145	1	**0**	1	1
199	**1**	0	1	1
200	**0**	0	0	1
219	0	**0**	0	1
220	0	**1**	1	1
269	0	**1**	1	1
270	0	**0**	0	1
299	**0**	0	0	1
300	**1**	0	0	0
334	1	**0**	0	0
335	1	**1**	0	0
399	**1**	1	0	0
400	**0**	1	1	0
454	0	**1**	1	0
455	0	**0**	0	0
474	0	**0**	0	0
475	0	**1**	1	0
499	**0**	1	1	0
500	**1**	1	1	1
599	**1**	1	1	1
600	**0**	1	1	1




CLK vs. Time (ps)
1.25
1.00
0.75
0.50
0.25
0.00
CLK
0
200
400
600
Time (ps)

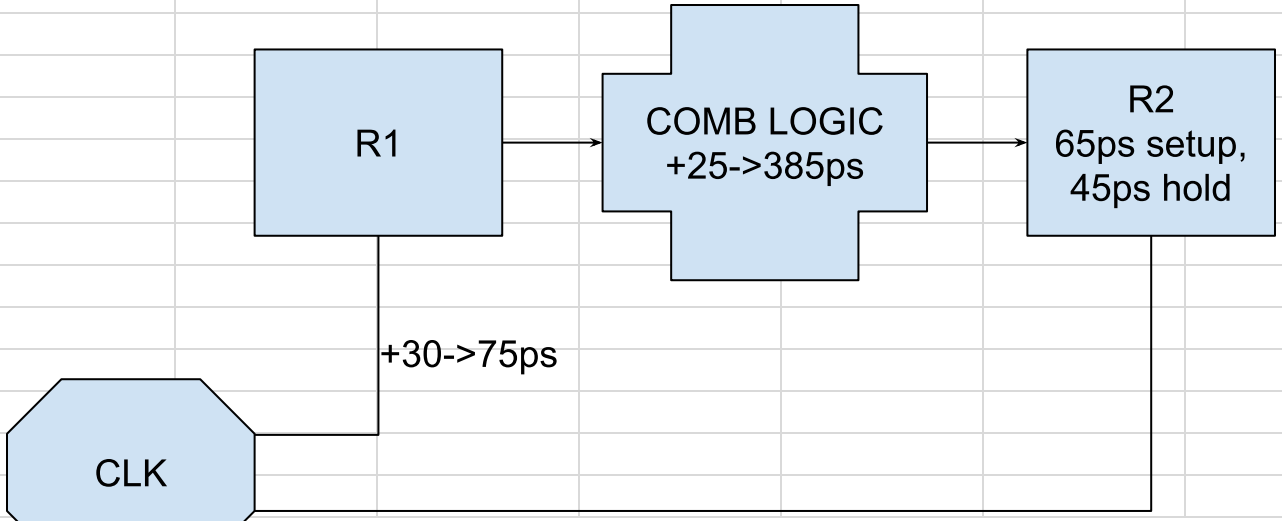

New D Value
1.25
1.00
0.75
0.50
0.25
0.00
Value
0
200
400
600
Time (ps)


QM vs. Time (ps)
1.25
1.00
0.75
0.50
0.25
0.00
QM
0
200
400
600



A change in D when the clock is high is not caught by the 'master', and so therefore is not observed by the 'slave'. The 'slave' observes the 'master' latch only, I believe?

3B

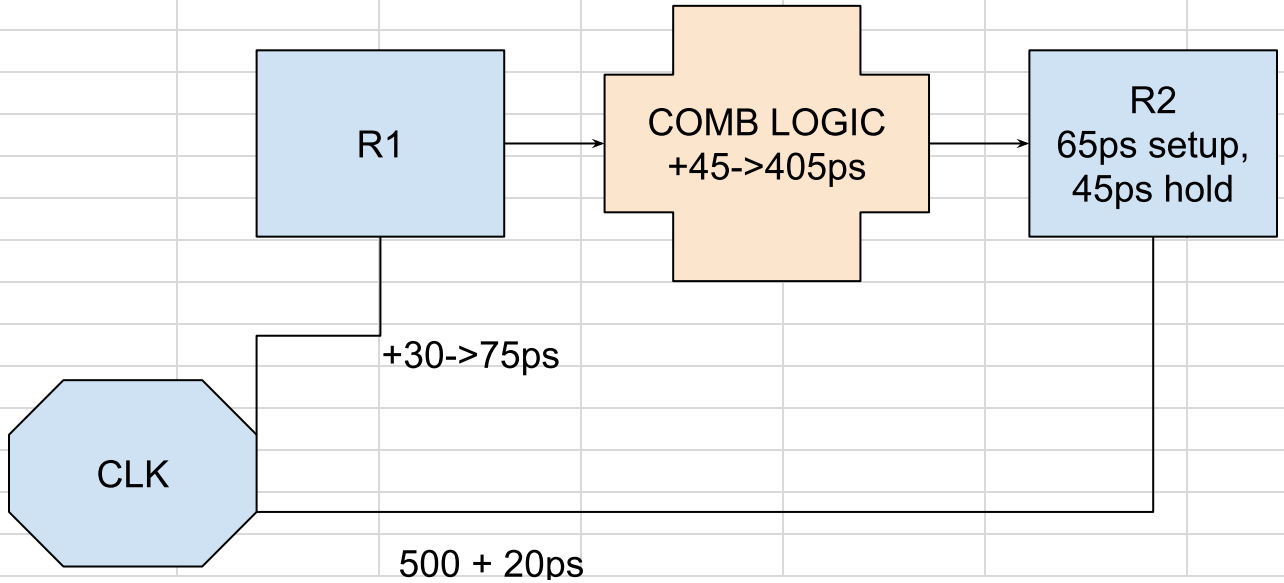




500 + 20ps

Maximum delay of data getting to R2:			It is possible that data would be output 5ps after R2 is ready.
75+385	equals	460	
Minimum time needed for R2 to be ready to accept data:			
500+20-65	equals	455	
Minimum delay of data getting to R2:			
30+25	equals	55	Data could arrive too soon and overwrite R2's inputs
Maximum time R2 needs data from previous clock:			
20+45	equals	65	

Added delay buffer to COMB LOGIC



Maximum delay of data getting to R2:				It is possible that data would be output 25ps after R2 is ready.		
75+405	equals	480				
Minimum time needed for R2 to be ready to accept data:						
500+20-65	equals	455				
Minimum delay of data getting to R2:				Data will arrive with a 10ps buffer. Hold time error fixed		
30+45	equals	75				
Maximum time R2 needs data from previous clock:						
20+45	equals	65				
Hold time errors are not affected by clock time? The error here was fixed, though worsening the set-up error, by increasing the time of the combinational logic block?						